

Probability Lecture 5

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Random Variables

- A random variable (r.v.) associates a value to every possible outcome
- Mathematically: A function from the sample space Ω to the real numbers
- It can take discrete or continuous values
- Notation: random variable X , numerical value x
- We can have several random variables defined on the same sample space
- A function of one or several random variables is also a random variable

Probability Mass Function (PMF) of a discrete r.v. X

- It is the "probability law" or a "probability distribution" of X
- If we fix some x then " $X = x$ " is an event

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega \text{ s.t. } X(\omega) = x\})$$

- $p_X(x) \geq 0 \quad \sum_x p_X(x) = 1$

Types of PMFs

Bernoulli

$$X = \begin{cases} 1, \\ 0 \end{cases}$$

$$p_X(0) = 1 - p$$

$$p_X(1) = p$$

Models a trial that results in success/ failure

Indicator r.v. of an event

$I_A = 1$ iff A occurs

$$p_{I_A}(1) = P(I_A = 1) = P(A)$$

Discrete Uniform

- **Parameters:** integers $a, b \quad a \leq b$
- **Experiment:** Pick one of $a, a + 1, \dots, b$ at random; all equally likely
- **Sample Space:** $\{a, a+1, \dots, b\}$
- **Random Variable:** $X : X(\omega) = \omega$
- **Model of:** complete ignorance

Binomial

- **Experiment:** n independent tosses of a coin with $P(\text{Heads}) = p$
- **Sample Space:** Set of sequences of H and T of length n
- **Random Variable:** X : number of heads observed
- **Model of:** number of successes in a given number of independent trials

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Geometric

- **Experiment:** infinitely many independent tosses of a coin; $P(\text{Heads}) = p$
- **Sample Space:** Set of infinite sequences of H and t
- **Random Variable:** X : number of tosses until the first heads
- **Model of:** waiting times; number of trials until a success

$$p_X(k) = P(X = k) = P(T \dots TH) = (1-p)^{k-1} p$$

$$P(\text{no Heads ever}) \leq P(TT \dots T) = (1-p)^k$$
$$\lim_{k \rightarrow \infty} (1-p)^k = 0$$

Expectation/ mean of a random variable

Definition: $E[X] = \sum_x x p_X(x)$

Bernoulli

If X is the indicator of an event A , $X = I_A$
 $X = 1$ iff A occurs

$$p = P(A), \quad E[I_A] = P(A)$$

Uniform r.v.

Remember $p_X(x) = \frac{1}{n+1}$

$$E[X] = 0 \frac{1}{n+1} + 1 \frac{1}{n+1} + \dots + n \frac{1}{n+1} = \frac{1}{n+1} \frac{n(n+1)}{2} = \frac{n}{2}$$

Elementary properties of Expectations

- If $X \geq 0$ then $E[X] \geq 0$ for all w : $X(w) \geq 0$
- If $a \leq X \leq b$, then $a \leq E[X] \leq b$ for all w : $a \leq X(w) \leq b$

$$E[X] = \sum_x x p_X(x) \geq \sum_x a p_X(x) = a \cdot 1 = a$$

- If c is a constant, $E[c] = c$

$$E[c] = c \cdot p(c) = c$$

Expected Value Rule for calculating $E[g(X)]$

Let X be a r.v. and let $Y = g(X)$

Averaging over y : $E[Y] = \sum_y y p_Y(y)$

Averaging over x : $E[Y] = E[g(X)] = \sum_x g(x) p_X(x)$

Proof

$$\begin{aligned} & \sum_Y \sum_{x:g(x)=y} g(x)p_X(x) \\ \Rightarrow & \sum_y \sum_{x:g(x)=y} yp_X(x) \\ \Rightarrow & \sum_y y \sum_{x:g(x)=y} p_X(x) \\ \Rightarrow & \sum_y yp_Y(y) = E[Y] \end{aligned}$$

Example

$$E[X^2] = \sum_x x^2 p_X(x)$$

This is not equal to $(E[X])^2$, this is true in most cases

Linearity of Expectation

$$E[aX + b] = aE[X] + b$$

Derivation

Based on the expected value rule

$$g(x) + ax + b, \quad Y = g(x)$$

$$\begin{aligned} E[Y] &= \sum_x g(x)p_X(x) \\ &= \sum_x (ax + b)p_X(x) \\ &= a \sum_x xp_X(x) + b \sum_x p_X(x) \\ &= aE[X] + b \end{aligned}$$

In this exceptional case $E[g(x)] = g(E[X])$